XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE 303
Electrical Machinery & Drives

COURSE OVERVIEW

By:
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Consultation Hour: by appointment

Programme Educational Objective (PEO)

PEO1	**Professionalism** Graduates will be capable of using problem-solving techniques and critical analysis to establish themselves as practicing professionals and demonstrate leadership skills and entrepreneurial spirit in the field of Electrical and Electronics Engineering or related fields.
PEO2	**Continuous Personal Development** Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for industrial and academic careers.
PEO3	**Societal Engagement** Graduates will contribute to sustainable development and the well-being of society.

Course Learning Outcome (CLO) and Programme Learning Outcome (PLO)

CLO1	Demonstrate the understanding of the first principles of electrical machines and drives.
CLO2	Investigate the expected performance of DC machines, synchronous machines and induction motor.
CLO3	Analyze various machines in the application of motor and generator such as synchronous machines, induction motors and DC motors.
PLO1	Engineering Knowledge - Apply knowledge of mathematics, science, electrical & electronic engineering fundamentals and engineering specialization principles to define and utilize engineering procedures, processes, systems and methodologies.
PLO4	Investigation - Conduct investigation into complex problems using research based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PLO2	Problem Analysis - Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

Timetable

- Lecture: Tuesday, 12pm – 2pm (A2#G11)
- Tutorial:
  - Group 1, Wednesday, 9 – 10am (A1#G02)
  - Group 2, Wednesday, 10 – 11am (A1#G02)
- Week: 1 – 15 (18 Sept 2023 – 29 Dec 2023

Course Assessment

Assessment	Detail	(%)
Continuous Assessment	Assignment (Group)	10
	Laboratory Reports (Group)	20
	Quiz (Individual)	20
Final Exam (2 hours) Answer All Questions	5 – 6 Questions	50
	Total	100

Grading

Exam Questions, Marking Scheme & Invigilation	
Special Requirements (For EAC approved Engineering Programmes)	• Multiple Choice Questions (MCQ) are not encouraged in the final exams. • **Students shall pass the final examination to pass the course, the threshold marks for final examination is 40.**

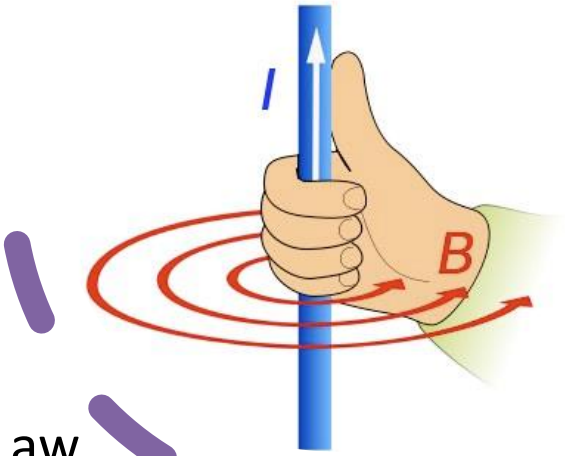
Marks	Grade	Grade Points	Description
85 – 100	A	4.00	Distinction
80 – 84	A-	3.70	Distinction
75 – 79	B+	3.30	Good
70 – 74	B	3.00	Good
65 – 69	B-	2.70	Good
60 – 64	C+	2.30	Pass
55 – 59	C	2.00	Pass
50 - 54	C-	1.70	Conditional Pass
45 – 49	D+	1.30	Fail
40 - 44	D	1.00	Fail
≤ 39	F	0.00	Fail

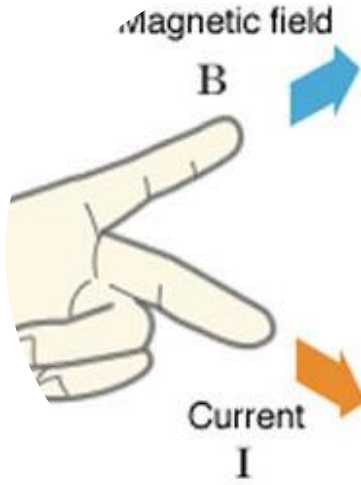
Grading scales and grades applicable to ALL intakes effective from Academic session 2017/09

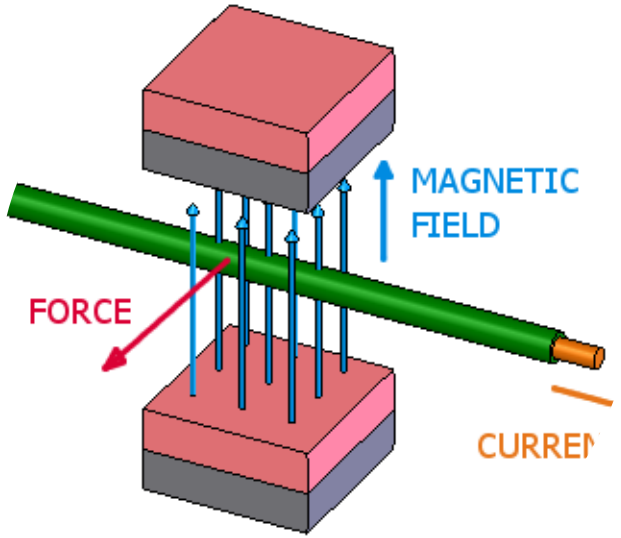
Course Content

Chapter 1: Introduction to Machinery Principles

- Rotational motion, Newton's Law, and power relationships
- The magnetic field, magnetic flux, magnetomotive and electromotive force
- Production of induced force on a wire
- Electrical machinery classification







Course Content

Chapter 2: DC Machine - Motor

- Introduction to DC machine
- Induced emf and speed of DC machine
- Types of DC motor
- Equivalent circuits and characteristics









Course Content

Chapter 3: DC Machine - Generator

- Types of DC generators
- Equivalent circuits and characteristics

Course Content

Chapter 4: AC Machine – Induction motor (IM)

- Introduction to AC machine
- 3-phase asynchronous /induction motor
- Basic construction, working principles, and equivalent circuit of IM
- Starting and control methods
- Modes of operation



Course Content

Chapter 5: AC Machine – Synchronous Generator

- Working principles and basic construction
- Frequency and generated voltage
- Equivalent circuit and phasor diagram
- Power and torque



Course Content

Chapter 6: AC Machine – Synchronous Motor

- Basic construction and working principles
- Starting method
- Equivalent circuit and phasor diagram
- Power and torque
- Effects of excitation
- Other applications







Course Content

Chapter 7: Single-phase and special purpose motor

- Universal motor
- Reluctance motor
- Stepper motor
- Servomotor

Group for labs and assignment work

- Form a group consisting 4 – 5 members.
- Fill in the names into the Excel sheet which I will send out later.
- We only have one Electrical Machine and Drive station in the lab, only one group can attempt the lab activity per slot, scheduling is necessary.
- Discuss among all group members and fill in the preferred slot in the Excel sheet, i.e. Monday 2 – 5 pm (3 hours recommended)

Attendance

- Attendance is required by the Ministry of Higher Education Malaysia.
- Sign the attendance sheet during class.
- If you cannot make it to class, sick leaves and other types of leaves should be applied 3 days prior/after, else you will be marked as absent.
- For two consecutive absences, you will be reported to Academic Affairs.

Course communication

- Email or Teams message
  - Requests for extensions, questions about course policies
  - Grading inquiries, requests for help with assignments
  - Sick leaves and other types of leaves
- Face-to-face
  - Class breaks
  - Email/message me to make an appointment for consultations

References



References

- Atif Iqbal, Shaikh Moinoddin, Bhimireddy Prathap Reddy. 'Electrical Machine Fundamentals with Numerical Simulation using MATLAB/SIMULINK'. 1st edition. Wiley. 2021.

Further Readings

- Bhattacharya S.K. ‘Electrical Machines’. 3rd ed. Tata McGraw-Hill. 2014.
- Wildi, T. ‘Electrical Machine, Drives and Power System’. Prentice-Hall. 2015.
- Stephen J. Chapman. ‘Electric Machinery Fundamentals’. 5th edition. McGraw-Hill. 2012.